ALTA AI Builders Fellowship

Week-1: Assignment

Task 1: Simple

1 Write a prompt to explain 'Machine Learning' in simple English with 2 examples

Ans:

Prompt-

> Explain Machine Learning in very simple English so that a beginner can understand it easily. Use clear and short sentences. Also include exactly 2 real-life examples to make the concept easy to understand. Avoid technical jargon and keep the explanation friendly and easy to read

Solution-

> Machine Learning is a way for computers to learn from data instead of being told exactly what to do.
>
> In simple words, it means teaching a computer by giving it examples, so it can find patterns and make decisions on its own.
>
> The more data the computer sees, the better it becomes at doing its task.
>
> **Example 1:**
> When you watch videos on YouTube, it suggests videos you may like. It learns from what you watch and shows similar content.
>
> **Example 2:**
> Email apps can detect spam messages. They learn from many emails and automatically move unwanted messages to the spam folder.

2 Write a prompt to generate a leave application email (student to teacher)

Ans:

Prompt-

> Write a polite and formal leave application email from a student to a teacher. The email should include the reason for leave, the number of days, and the dates of absence. Use

simple and respectful language. Keep the email short and clear. Also include a proper subject line and a closing thank-you note.

Solution-

Dear Sir/Madam,

I hope you are doing well.

I am writing to inform you that I will not be able to attend classes for 3 days, from 18 April 2026 to 20 April 2026, due to health issues. I kindly request you to grant me leave for these days.

I will make sure to complete all the missed work after I return.

Thank you for your understanding.

Yours sincerely,
Vishakha

Task 2: Moderate

1 Write a prompt to create a structured resume using your details

Ans:

Prompt-

Create a professional and structured resume using my details. Include sections like Name, Contact Information, Objective, Education, Skills, Projects, Experience (if any), Certifications, and Achievements. Format the resume neatly with clear headings and bullet points. Use simple and professional language. Ensure the resume is suitable for a [job role, e.g., Front-End Developer / Data Analyst]. Ask me for any missing details if needed before generating the resume.

Solution-

1. **Contact Information**
   - Phone number
   - Email
   - LinkedIn (if any)
   - GitHub/Portfolio (if any)

2. Objective (optional)

- Do you want a custom career objective or should I write one for you?

3. Education

- Degree (you mentioned BTech CSE ✓)
- College name (PES College of Engineering, Aurangabad ✓)
- Year of graduation
- CGPA/percentage (optional)

4. Skills

- Programming languages (Python, Java, etc.)
- Tools/Technologies (ML, AI, NLP, TensorFlow, etc.)
- Web/Other skills

5. Projects
(You already have some, confirm/add details)

- Deepfake Detection System
- SMS & Email Spam Classifier
- News AI Agents
- Gesture Volume Control
- Any others?
  → For each: short description + technologies used

6. Experience (if any)

- Internships (You mentioned NIELIT ✓ — add duration + work done)

7. Certifications
(You mentioned some ✓)

- Google Cloud (Prompt Design & Vertex AI)
- Data Science Workshop (Braino Vision)
- Any others?

8. Achievements
(You have many ✓, confirm if you want all included)

- 3rd Prize – Code Fest Python
- Winner – Technical Paper Presentation
- GSSoC Contributor
- Others?

2 Write a prompt to compare AI vs ML vs Generative AI in table format

Ans:

Prompt-

> Compare Artificial Intelligence, Machine Learning, and Generative AI in a structured table format. Include key differences such as definition, scope, working method, examples, and use cases. Use simple and easy-to-understand language. Keep the table clean, well-organized, and beginner-friendly.

Solution-

Aspect	**Artificial Intelligence (AI)**	**Machine Learning (ML)**	**Generative AI (Gen AI)**
Definition	AI is the broad field of making machines act smart like humans.	ML is a part of AI where machines learn from data.	Gen AI is a part of ML that creates new content.
Scope	Very wide (includes ML, robotics, vision, etc.)	Narrower than AI (focuses on learning from data)	More specific (focuses on content creation)
Working Method	Uses rules, logic, and learning methods	Learns patterns from data and improves over time	Learns patterns and generates new text, images, etc.
Examples	Voice assistants, self-driving cars	Email spam detection, recommendation systems	Chatbots, image generation tools
Use Cases	Automation, robotics, decision-making	Prediction, classification, data analysis	Content creation, design, text generation

Task 3: Hard

1 Create one detailed prompt using Role + Task + Context + Format + Constraints

Ans:

Prompt-

Role:
You are an expert career coach and resume writer specializing in technology roles.

Task:
Create a professional, ATS-friendly resume for a student applying for an AI Engineer role.

Context:
The student is currently pursuing a BTech in Computer Science Engineering and has knowledge of Python, Machine Learning, and basic web development. They have worked on projects like a Deepfake Detection System, Spam Classifier, and AI-based applications. They also have internship experience and certifications in AI/ML-related fields.

Format:

- Use clear section headings: Name, Contact Information, Objective, Education, Skills, Projects, Experience, Certifications, Achievements
- Present information using bullet points
- Keep the layout clean and professional
- Use simple and formal language

Constraints:

- Keep the resume within 1 page
- Avoid unnecessary technical jargon
- Ensure clarity and readability
- Highlight AI/ML skills and projects prominently
- Do not include false or assumed information

2.Example task: Explain any topic + give examples + summary + table

Ans:

Prompt-

Create a full answer with explanation, examples, table, and summary using one topic for my assignment that is **"Using a smartphone app (YouTube, Google, or WhatsApp)"**

Solution-

Topic: Machine Learning (Using Smartphone Apps)

Explanation

Machine Learning is a method where computers learn from data and improve automatically.

In simple words, instead of telling a computer every step, we give it data. The computer studies this data, finds patterns, and makes decisions.

Smartphone apps like YouTube, Google, and WhatsApp use Machine Learning to make our experience better and smarter.

Real-Life Examples

1. YouTube App
When you watch videos on YouTube, it suggests new videos based on what you like. It learns from your watch history and shows similar content.

2. Google Search
When you type something in Google, it gives suggestions and shows the most relevant results. It learns from what people search and click

Table: Machine Learning in Smartphone Apps

App	What It Does	How Machine Learning Helps
YouTube	Recommends videos	Learns from watch history and preferences
Google	Shows search results	Learns from user searches and behavior
WhatsApp	Filters spam messages	Detects unusual or unwanted messages

Summary

Machine Learning is used in everyday smartphone apps to make them smarter. Apps like YouTube, Google, and WhatsApp learn from user behavior and provide better suggestions, results, and security. This makes our daily digital experience faster and more personalized.